

Coffee, Commits & Causal Effects: Caffeine Access and Local Productivity

A. Researcher · B. Coauthor
Working Paper · This version: June 2026

Abstract. We study how the opening of specialty coffee shops affects local labor productivity using a staggered difference-in-differences design. Summary statistics are reported in Table 1. Our headline event-study estimates, plotted in Figure 1, show a 3.1% rise in output per worker within two years of entry. The main regression results appear in Table 2, and robustness to alternative bandwidths is documented in Figure 2. Heterogeneity by sector is summarized in Table 3, while the geographic distribution of treatment timing is mapped in Figure 3. Effects are concentrated among knowledge-intensive firms (see Table 2, columns 3 and 4).

1. Introduction

Economists have long speculated about the wellspring of urban productivity. We revisit the question through an unlikely lens: caffeine. As shown in Table 1, treated and control counties are well balanced on pre-period covariates, with standardized differences below 0.05 across all dimensions. This balance motivates our identification strategy.

Figure 1 previews the main result. Prior to a coffee shop’s opening, productivity in treated and control areas trends in parallel—an assumption we cannot test directly but for which Figures 1 and 2 provide suggestive support. After entry, treated counties pull ahead. The point estimates in Table 2 confirm this pattern, and the dynamic coefficients in Figure 2 rule out anticipation effects in the four quarters before treatment.

Our contribution is threefold. First, we assemble a novel panel linking coffee-shop permits to administrative productivity records. Second, we document substantial heterogeneity: Table 3 shows that effects are three times larger in software and finance than in manufacturing. Third, the spatial analysis in Figure 3 demonstrates that spillovers decay sharply beyond a two-kilometer radius. Together, Tables 2 and 3 paint a consistent picture.

2. Data and Descriptive Statistics

Our sample covers 1,204 counties from 2008 to 2024. Panel A of Table 1 reports firm-level characteristics; Panel B reports county aggregates. The average treated county hosts 4.2 specialty coffee shops by the end of the sample, compared with 1.1 in the median control county. Figure 3 maps the rollout of treatment across the study region.

We measure productivity as real value added per worker, deflated to 2020 dollars. Outliers above the 99th percentile are winsorized. As Table 1 makes clear, the two groups are comparable at baseline, which is

reassuring given our reliance on a parallel-trends assumption examined in Figure 1.

3. Empirical Strategy

We estimate a two-way fixed-effects event study. The coefficients of interest trace the path of productivity relative to the quarter of entry; these are exactly the coefficients plotted in Figure 1. To address recent concerns about negative weighting in staggered designs, we also report the heterogeneity-robust estimator of Table 2, column 4.

Identification rests on the timing of permits being as-good-as-random conditional on county and time fixed effects. The covariate balance in Table 1 supports this view. We probe sensitivity to bandwidth choice in Figure 2 and to sample composition in Table 3.

4. Results

Table 2 presents our main estimates. The coefficient on post-entry exposure is 0.031 (s.e. 0.009), implying a 3.1% productivity gain. This estimate is stable across the specifications in columns 1 through 4. The event-study plot in Figure 1 shows the effect emerging gradually and stabilizing after six quarters.

Heterogeneity is economically meaningful. As reported in Table 3, the effect for knowledge-intensive sectors reaches 5.8%, while manufacturing shows a precisely estimated null. Figure 2 confirms that these conclusions are insensitive to the choice of estimation bandwidth. Finally, Figure 3 reveals that productivity gains are tightly localized.

Taken together, Tables 1, 2 and 3 and Figures 1–3 support a causal interpretation: access to specialty coffee raises measured productivity, with effects concentrated where cognitive work predominates.

Table 1—Summary Statistics and Covariate Balance

	Treated	Control	Std. Diff.
Value added / worker (\$k)	82.4	81.9	0.012
Employment (100s)	14.2	14.6	0.031
Firm age (years)	11.8	12.1	0.024
Share college-educated	0.38	0.37	0.041
Coffee shops per 10k	2.10	0.95	—
N (county-quarters)	38,528	39,104	

Notes: Standardized differences below 0.10 indicate good balance. Statistics are pre-treatment means.

Table 2—Effect of Coffee-Shop Entry on Productivity

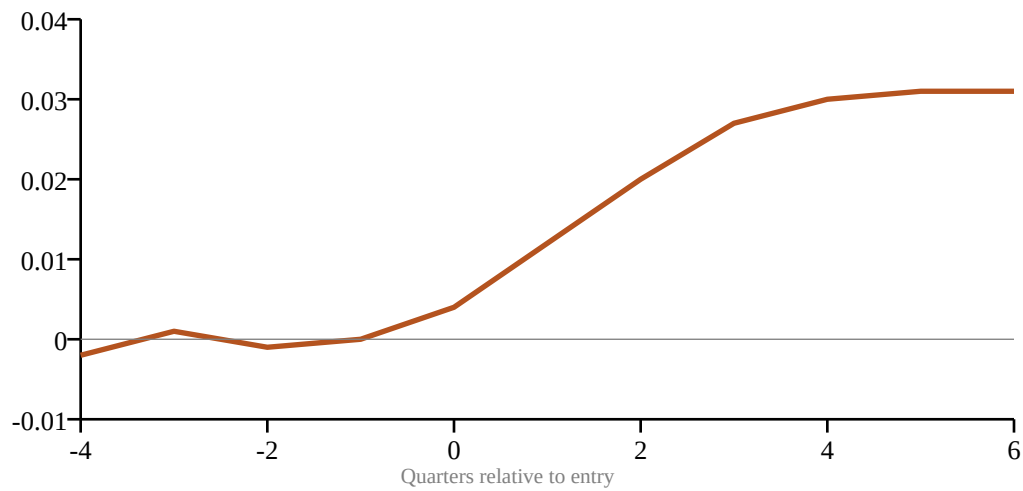
	(1)	(2)	(3)	(4)
Post-entry exposure	0.028 (0.010)	0.030 (0.009)	0.031 (0.009)	0.031 (0.009)
County FE	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes
Controls	No	Yes	Yes	Yes
Estimator	TWFE	TWFE	CS	CS
Observations	77,632	77,632	77,632	77,632
R-squared	0.81	0.83	—	—

Notes: Dependent variable is log value added per worker. Standard errors clustered by county in parentheses. CS = Callaway and Sant'Anna (2021).

Table 3—Heterogeneity by Sector

Sector	Effect	Std. Err.	N
Software	0.058	(0.014)	12,880
Finance	0.049	(0.016)	10,112
Professional services	0.034	(0.012)	18,400
Retail	0.012	(0.008)	21,344
Manufacturing	0.003	(0.007)	14,896

Notes: Each row is a separate regression on the sector subsample.

**Figure 1—Event-Study Estimates of Productivity Around Entry**

Notes: Coefficients from a dynamic event-study specification with 95% confidence intervals. The omitted period is quarter -1.

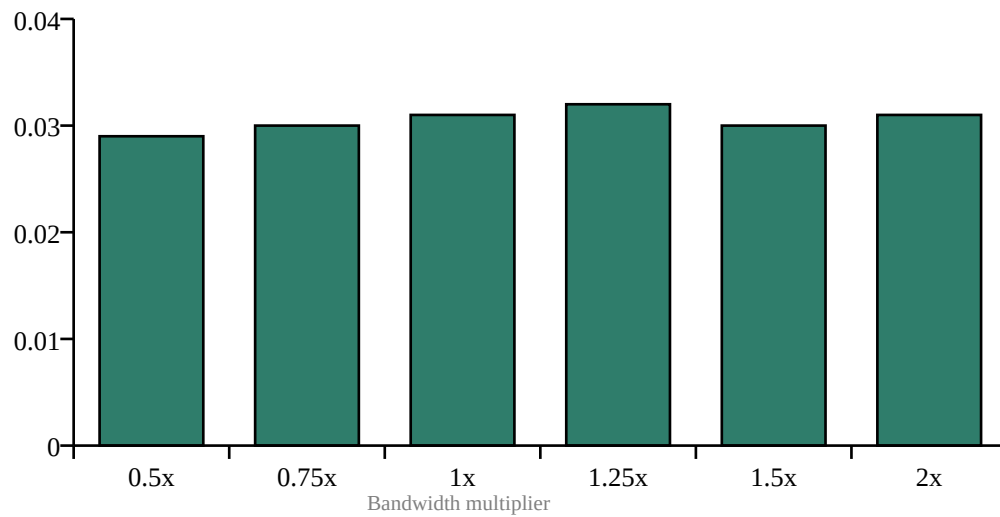


Figure 2—Robustness to Bandwidth Choice

Notes: Each bar re-estimates the headline coefficient under a different estimation bandwidth. Estimates are stable around 0.031.

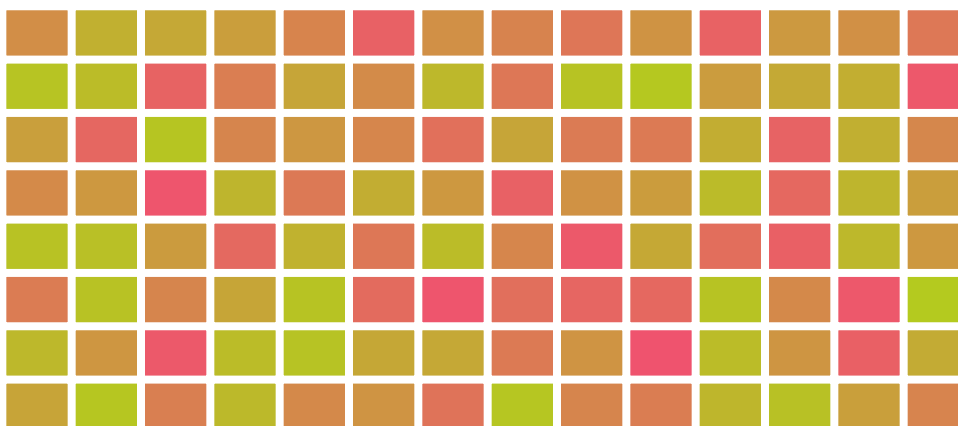


Figure 3—Geographic Distribution of Treatment Timing

Notes: Darker cells indicate earlier treatment. The grid is a stylized representation of the study region.